

Maker Portfolio

Teadora Gildengers

Fairy Ring, Spring 2024



“Fairy Ring”

During my undergraduate degree at CMU I completed a minor in Soft Technologies through IDeATe, an interdisciplinary department focused on maker projects. Fairy Ring is a connected ring of inflatable mushrooms that can be arranged for a group to sit inside, developed as a two-person team in my coursework. We rendered a mushroom stem in CAD and cut out the segments as templates on a laser cutter available at one of CMU’s makerspaces, which made executing the design much more effective. To this day Fairy Wing is one of the maker projects I am most proud of: it’s incredibly tactile, interactive, and came out just like I imagined it.

My writeup on Fairy Wing is available at

<https://sites.google.com/andrew.cmu.edu/53-330a-ideate-inflatables/pushpull-spaces/bryce-teadora-fairy-ring>

Poke-turnip, Spring 2024



“Poke-turnip”

“Poke-turnip” is a 20-foot tall installation I made as the final project of my Inflatables coursework in Spring 2024. I worked in a team of 5 to develop CAD, prototype modular designs, dye fabric, assemble the installation, and install it for a public showcase with 200+ attendees. Inspired by my work on “Fairy Ring”, we used a laser cutter to cut and label every piece of the inflatable. This saved us significant time, improved accuracy, and made assembly much easier. During the exhibition we connected the leaves to a pulley so participants could move them up and down. To this day, “Poke-turnip” is the largest project I’ve ever made, and I’m proud of how I combined my creativity and engineering skills in its construction.





During Summer 2023 I [developed code](#) in a two-person team for the CMU Textile Lab's computerized knitting machine that could knit a well-fitting sweater in one piece. Throughout the project, I learned how to operate and repair the machine, read and write the machine's custom software, and used my design skills to improve the garment fit. Our work was presented at [SCF 2023](#), the ACM Symposium on Computational Fabrication.